

# The evolution of aviation network: Global airport connectivity index 2006–2016



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## ABSTRACT

This paper studies the evolution of global air transport network over the period 2006–2016. A Global Airport Connectivity Index (GACI) is proposed, combining degree, closeness and eigenvector topological indicators and two new volumetric indicators. We find faster passenger growth for larger airports that have improved in GACI. While America and North Asia focused on regional hub developments, serving short- to medium-haul routes, West Europe, South East Asia and Middle East focused on developing their major airports into global hubs. Our analysis shows that increasing GACI improves an airport's competitive position and gives rise to increased influence over other airports.

## 1. Introduction

Air transport has both contributed to and benefited from increased participation in the global economy that has helped many cities in the world become deeply connected to close trading partners. The direct economic benefits from aviation are palpable. It is estimated that the air transport sector directly employed 10 million people and generated US\$3.0 trillion in value-added output globally (ATAG, 2018). Continued demand for air transport, together with ongoing opening up of the skies, provided incentive for airport-cities to further expand their connectivity. The number of unique city-pair connections exceeded 18,400 in 2016, a 4% increase from the previous year and a near doubling of the connections 20 years ago (IATA, 2017). The measure of connectivity has become a key indicator of the economic importance of a city in the global air transport network (IATA, 2019; Arvis and Shepherd, 2013). Higher level of air connectivity is said to have greater potential in creating advantages for employment growth, delivering economic and social benefits to the area (e.g., Irwin and Kasarda, 1991; Pearce, 2007; Bel and Fageda, 2008; Zhang, 2012; Florida et al., 2015). Consequently, there is a significant interest to analyse air transport networks in terms of connections. The study of air connectivity is also an important tool to assist stakeholders' forecast and planning. It helps benchmark and monitor the network performance against that of other airports or airlines (Burghouwt and Redondi, 2013). Industry leaders or policy makers can make use of the study findings to identify major competition and cooperation opportunities (Burghouwt and Veldhuis, 2006). The results may, for instance, provide a clearer picture of destinations and airport requirements needed to attract airlines to operate new international routes.

Understanding the topology of air transport network is important for two primary reasons. First, the global connectivity by air is growing larger and more complex. Connectivity represents a momentous proxy in understanding passengers' route choice, including

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their “self-hubbing” (“self-connecting”) strategies (Malighetti et al., 2008; Suau-Sanchez et al., 2016; Zeigler et al., 2017). The knowledge of emergent phenomena, such as hub airport capacity as a competition barrier, the rise of middle-eastern air carriers and the burgeoning markets of Asia, may be improved by looking at patterns of airport connectivity in a quantitative framework, where passenger flows, direct and indirect linkages between airports, and airports’ regional importance are explicitly taken into consideration. Second, being a physical network, the air transport network can be viewed as a topological entity. As such, it possesses measurable network properties, which can help to statistically explain how major forces are reshaping global passenger flows, for example, the significant shift in air passenger demands that has been occurring after the global financial crisis, most notably toward emerging economies (Goyal and Negi, 2014; Wong et al., 2019a, 2019b). Furthermore, airport position in the air transport network has substantial implications across many areas of a country’s economy, especially those heavily reliant on air transportation as a source of growth. The key importance is that an airport’s ability to effectively connect to others affect not only in shaping patterns of air transport network, but also in determining relative centralities between airports, and ultimately comparative development.

Globally, the air transport network involves around fifty very large, interconnected airports (i.e. global hubs), which collectively account for around 34% of the world’s seat capacity. Despite their central roles, how the connectivity around these hubs arises in the first place remains largely unexplained. In other words, why do certain airports become more connected and systemically important than others? Air transport network, in its complex network representation, has been extensively studied and a lot has been learned about its topological properties and about how the network, or part of it, evolves. For example, Bounova (2009) studied the topological evolution of US air network and found that the prototypical low-cost carrier (Southwest Airlines) network was becoming more centralized, gradually transitioning towards the hub-and-spoke network model. Dai et al. (2018) investigated the topological changes of the Southeast Asian air transport network from a complex network perspective over a 23-year period and found that it exhibited an intensifying disassortative mixing pattern: cities with many connections (i.e. high-degree nodes) such as Singapore and Bangkok are on average connected with cities with few connections (i.e. low-degree neighbouring nodes). This suggests an increasing dependence of small cities on hub-and-spoke configuration for better accessibility. Furthermore, Rocha (2017) reviewed the available literature and summarised the results of the empirical analysis of evolving air transport networks in several different markets, including China, Brazil, US, and Europe. All his studied airport networks exhibited a positive-skewed distribution of degree, suggesting the coexistence of a few hub airports with many route connections and majority of airports with only a few routes. Other connectivity applications include airport classification for a certain market, such as the US, Europe, or the Southeast Asia–North America market (Rodríguez-Déniz et al., 2013; Paleari et al., 2010; Suau-Sanchez et al., 2015; Choi et al., 2019; Zhu et al., 2019). There are also studies focusing on national air transport networks, such as Wang et al. (2011), Bing (2014), Shao and Sun (2016), Wang et al. (2016), and Zhang et al. (2017) for China, Hua et al. (2010), Cheung and Gunes (2012) and Jia et al. (2014) for United States, Oliveira et al. (2016) and Couto et al. (2015) for Brazil, Hossain et al. (2013) for Australia, and Bagler (2008) for India. The results from these studies indicate that the importance of an airport within the air transport network and the potential at which an airport is passed or by-passed when connecting passengers from an origin to a destination are linked concepts, both closely related to the measures of airport centrality.

Furthermore, air transport network is considered not only by its topology, but also by the dynamics of weight (i.e. passenger flows) taking place along the routes. For instance, the heterogeneity in the strength of connections is very important in understanding air transport markets. By considering all the connections with equal weight, the topological approach might underestimate the impact of passenger flow heterogeneity when analysing airport centrality. For instance, the statistical properties of the weighted air transport network differ markedly in some critical parameters from those found by the topological approach. Specifically, the strength distribution is heavily positive-skewed, indicating that the passenger flows are concentrated on a few routes and a vast majority of routes have small weights. In the last decade, more than half of the world’s passenger seat capacity were covered only by 2.4% of airports, although 6% of the airports operated half of the world’s routes. This is also apparent in the work of Guimerà et al. (2005), who found that the world’s airport-cities with more connections were not always the most central in the network. Despite this, globally central airports typically have many connections, but only a few of them have very large passenger flows.

Several studies have quantified the evolving properties of air transport networks in different countries or world regions using a range of measures in the topological representations with traffic information considered. For instance, Wandelt and Sun (2015) analysed the evolution of the world’s air transport networks at country level over a 12-year period and identified countries and flight connections that were topologically critical in an unweighted network and functionally critical in a weighted network. Recently, they also compared the passenger traffic variations, together with the topological properties, of eight domestic air transport networks over the same 12-year period (Wandelt et al., 2019). They showed that most networks exhibited clear seasonal traffic patterns, and all networks exhibited 95% similarity between two consecutive months. Using different kinds of measures, Wang et al. (2016) applied market competition and concentration indexes to analyse the route development of Chinese air transport network in response to deregulation, competition, and airline reorganisation over an 18-year period. Finally, Allroggen et al. (2015) proposed an air connectivity metric and then evaluated the link quality and hub centrality at the world’s airports in terms of frequency, directness and market accessibility. The impact on global connectivity of an airport in response to network changes was evaluated at the route level and limited to direct or one-stop connections. While these studies employed several different centrality indicators and network metrics as the proxies of different properties of air transport networks, they largely neglected the issue of whether those measures were able to (i) reflect the connect-ability exemplified by the passenger flows, and (ii) explain how an airport’s connectedness has increased or decreased over time.

To provide a detailed snapshot of how airports participate in the global passenger flows, this paper has created a Global Airport Connectivity Index (GACI) by combining a few selected centrality indicators in a concise way so as to quantify a range of fundamental properties of the weighted network. GACI provides a benchmark that is able to compare the importance of individual airports at a global scale across time. By examining GACI changes, one is able to identify the relevant forces driving global passenger flows. For

some airports, flow intensity, which is the volume of passenger traffics relative to their connectivity, is significant and growing. GACI is also able to assess the impact of increased connectivity to passenger flows, and study how an airport's connections with certain airports can amplify the benefits of passenger flows and propagate onward through subsequent links. This network effect is considered in this paper.

The paper is organized as follows. In [Section 2](#) we first briefly describe the set of relevant topological and weighted network measures, and then discuss the data collection and preliminary findings of the changes in the global air transport network. Both the individual indicators for airport connectivity and the overall airport centrality index (i.e. GACI) performance from 2006 to 2016 will be presented in [Sections 3 and 4](#), respectively. In [Section 5](#), we examine how GACI could help airports and airlines management as well as policy makers to benchmark the airports' position in the global air transport network. Finally, [Section 6](#) contains concluding remarks.

## 2. Data and metrics

### 2.1. Data preparation

Our flight data comes from the Official Airline Guide database ([OAG, 2019](#)), which offers a comprehensive coverage of approximately 1000 airline schedules. Specifically, route segment data is used in this study. It gives weekly passenger seat capacity of all the scheduled flight services around the world. The data set covers over 3500 airports and 60,000 unidirectional routes worldwide over an 11-year period between 2006 and 2016. Segment information is then aggregated over all the flights between every connected airport pair, in each year. Other flight information recorded in the segment includes origin and destination airports, operating carriers, and flight frequency. The data set also contains airport location information on city, country, and region.

To focus on scheduled commercial air transport services, four types of airports are excluded in this study: (i) Military airports, such as air force base; (ii) Airports for private jets, charters, seasonal or seaplanes only; (iii) Heliports; and (iv) Airports that were closed between 2006 and 2016. A summary of the air transport networks in 2006, 2011 and 2016 is given in [Table 1](#). While the number of airports has a moderate increase of around 6%, the number of direct flight routes and total capacity have both increased substantially by more than 25% and 44%, respectively, over the 11-year period. Furthermore, density has increased, indicating that more direct linkages are formed between smaller airports. This is in part also supported by the reduction in diameter of the network and the gain in average degree over the same period.

### 2.2. Metrics formulations

The air transport network is constructed from airports and flight routes. It consists of nodes representing the airports, and links that connect the nodes. Two airports are connected in a given year if they have at least one scheduled flight between them. We use a graph  $G(V_N, E_L)$  to describe the air transport network, where  $V_N$  is a set of  $N$  airport nodes interconnected by a set of  $L$  links,  $E_L$ . Each link  $(i, j)$  is associated with a link weight  $w_{ij}$ , which represents the total number of air passengers on that link. All the links are collected and their aggregated weights are computed within one year into a network snapshot for each of the 11 years. Hence, the snapshot of the network in a given year is represented by the weight adjacent matrix,  $W = [w_{ij}]$ . Furthermore, the topology of the network is represented by its non-weighted adjacent matrix  $A = [a_{ij}]$ , where  $a_{ij} = 1$  if node  $i$  and node  $j$  are connected, and 0 otherwise.

The quality of links can be summarized by five network centrality indicators: degree, volumetric-based flow betweenness, closeness, eigenvector, and regional importance. Together, these indicators capture distinct aspects of what it means to be central in the air transport network for individual airports. In particular, the flow-betweenness and regional-importance are uniquely developed to reflect the short-haul, flow-centric characteristic in air travels.

Their definitions are as follows:

1. **Degree**  $d_i$  of an airport node  $n_i$  is the number of other nodes to which  $n_i$  is connected, that is,  $d_i = \sum_{j=1}^N a_{ij}$ . The degree distribution over time gives an idea of how the whole network topology is changing ([Burghouwt and Hakfoort, 2001](#); [Wandelt and Sun, 2015](#); [Cong et al., 2016](#)), whereas exploring the association patterns among the node-degrees gives insights into the network structure. Let  $D_r$  be the number of nodes with degree  $r$ . The node degree distribution  $P(r)$  is:

$$P(r) = \frac{D_r}{N} \quad (1)$$

**Table 1**

Passenger Seat Capacity and Major Properties of Global Air Transport Network, 2006–2016.

| Year | Nodes | Routes (direct) | Total Capacity (1 0 <sup>6</sup> ) | Density (1 0 <sup>-3</sup> ) | Diameter (routes) | Degree Centrality ( $\bar{d}_i$ ) |
|------|-------|-----------------|------------------------------------|------------------------------|-------------------|-----------------------------------|
| 2006 | 3291  | 58,678          | 7,599.9                            | 5.42                         | 9                 | 17.83                             |
| 2011 | 3455  | 67,662          | 9,041.1                            | 5.67                         | 8                 | 19.58                             |
| 2016 | 3510  | 73,510          | 10,969.5                           | 5.97                         | 7                 | 20.94                             |

Source: OAG data with authors' calculation.

2. **Flow Betweenness  $F_b(i)$** : The original shortest-path betweenness centrality  $C_b(i)$  of a node  $i \in V_N$  is defined as the relative fraction of shortest paths passing through  $i$  (Freeman, 1977; Liu et al., 2011). The betweenness is based on the assumption that travellers would choose the shortest path when they travel from one point to another (Guimerà et al., 2005; Malighetti et al., 2008; Lin and Ban, 2014). The path length is the number of links connecting the sequence of nodes in the path. Shortest paths between a pair of origin and destination nodes are the smallest number of links (in  $E_L$ ) between them. Let  $\sigma_{jg}$  be the number of shortest paths between  $j$  and  $g$  passing through  $i$ , for  $j, g \in V_N$ .  $C_b(i)$  is calculated as:

$$C_b(i) = \frac{1}{N_B} \sum_{j \neq g \neq i \in V_N} \frac{\sigma_{jg}(i)}{\sigma_{jg}} \quad (2)$$

where  $N_B = \frac{1}{2}N(N-1)$  is a normalizing constant.

Such definition of betweenness centrality accounts only the shortest paths, while ignoring those other paths which contain links that can be important in the air transport network. Air passenger movement is typically planned from a specific origin-to-destination and may not just travel on the shortest path only. However, without the actual passengers' itinerary records, the shortest path betweenness does not relate well with the passenger flows. As our OAG dataset covers only scheduled flight seat capacity data, the absence of complete itinerary transit information, which includes both single and multiple itineraries with multiple stops, makes it difficult to assess the actual passenger flows through the network. With such data being unavailable, we adopt the flow betweenness centrality introduced by Brandes and Fleischer (2005) and Newman (2005), with modification to account for the intensity of transfer passenger flows between two airports. As such, an air transport network can be regarded as an electrical network with links acting as resistors and nodes the junctions between resistors. The current-flow betweenness of a node is proportional to the amount of "current" passing through the node. Each link is assigned a positive resistance value inversely proportional to the intensity of the relationship between the source and destination airports. This value can be approximated by the following conditional probability distribution:

$$P(w_{ij} > x | d_i, d_j, s_i, s_j) = e^{-\frac{x}{\langle w_{ij} \rangle}} \quad (3)$$

where  $\langle w_{ij} \rangle = \sqrt{\frac{s_i s_j}{d_i d_j}}$  is the geometric mean of the average link weights of nodes  $i$  and  $j$ . We assume the link intensity to be proportional to the average weight of its two end nodes. The use of geometric mean log-transformed the ranges being averaged, so that it is less dominated by the larger value of the two ( $s/d$ ).

The flow betweenness  $F_b(i)$  of an airport node  $i$  is averaged over all origin–destination pairs in the weighted air transport network:

$$F_b(i) = \frac{1}{N_B} \sum_{j \neq g \neq i \in V_N} \tau_{jg}(i) \quad (4)$$

where  $\tau_{jg}(i)$  is the passenger flow through node  $i$  between origin  $j$  and destination  $g$ . Thus,  $F_b(i)$  measures the fraction of passenger flows passing through node  $i$  between all possible airport-pairs in the weighted network. As passenger flow in the network traverses many paths, not all the connections are of equal weight. This is illustrated schematically in Fig. 1. The definition also indicates that the nodes with high flow betweenness connect mostly to network paths whose source and destination airports interact with high intensity. Compared with other connectivity measures, our intensity-based flow-betweenness is still based on topology network. However, it is evaluated for each airport-pair, which can provide some disaggregated information in probabilistic sense, such as route choices, from passengers' perspective. The limitation of this measure is that it is not able to give an assessment of airports' transfer connectivity managed by airlines or by passengers themselves (self-connecting), which depends at least partly on the airlines' business models, including their schedules to determine the number of potential flight connections that meet the connection time constraints of airports. Nevertheless, this measure may be used as a reference for the proportion of transfer passengers at an airport.

3. **Closeness  $C_c(i)$**  measures the accessibility of an airport in a network (Malighetti et al., 2008; Wang et al., 2011) or an airport proximity to the other nodes (Newman 2005; Mao et al. 2015). It measures the mean path length from a node to other nodes,

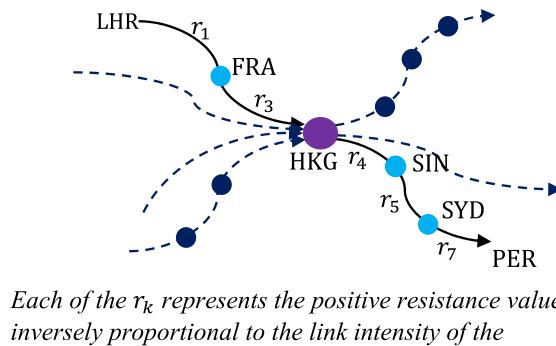


Fig. 1. A network configuration shows the case where airport HKG is contained in a given route from airport LHR to airport PER.

assuming the shortest path between two nodes. The length of a path from  $i$  to  $j$  is the number of links along the path. It is defined as:

$$C_c(i) = \frac{N-1}{\sum_{j \in V_N, j \neq i} \text{Dist}(i, j)} \quad (5)$$

where  $\text{Dist}(i, j)$  is the path length from  $i$  and  $j$ .  $C_c(i)$  measures the inverse of the average path length between node  $i$  and other nodes. This centrality indicator estimates indirectly the traffic flow that may have passed through an airport compared to other airports. An airport with lower mean path length may find that its passengers can reach their destinations with less stopovers than other airports.

4. **Eigenvector centrality**  $C_E(i)$  is based on the assumption that not all the connections are equal. It measures the importance of a node in the network by incorporating the importance of its connected neighbours (Alderighi et al., 2007). An airport having many connections do not necessarily have high eigenvector centrality, as the nodes to which it is connected might have lower eigenvector centrality than it has. To keep track of all the connected nodes, the weighted matrix  $W$  is used. Here, the weight is interpreted as connection strength. It is defined as:

$$C_E(i) = \frac{1}{\lambda} \sum_j W_{ij} C_E(j) \quad (6)$$

where  $\lambda \neq 0$  is the largest eigenvalue.

5. **Regional importance**  $C_R(i)$  is a quality indicator that utilizes the within-region connectivity and link intensity (given as  $\langle w_{ij} \rangle$  in Eq. (3)) together to quantify an airport's importance in its region. Separating airports into their regions creates a kind of community structure, in which the importance of each airport to its community can be measured. Guimerà et al. (2005) studied community structures in the air transportation network. Rather than regions, their community structures were based on finding groups of densely connected airport-cities. They established the roles of each airport-city based on its pattern of inter-community and within-community connections. Here,  $C_R(i)$  of a node  $i \in V_N$  is defined as the mean link intensity  $\langle \bar{w}_{ij} \rangle$  over all the connections between  $i$  and other nodes  $j$  inside the same region of  $i$ . An airport has a high  $C_R(i)$  when its connections inside the region exhibit high average intensity.

While many measures of centrality are available to characterize an airport's positions and route connections in air transport networks, we limit ourselves to these five centrality indicators which, we believe, are sufficient to cover most identifying characteristics of an air transport network. Some of the other common measures include clustering coefficient (Watts and Strogatz, 1998), passenger flux (Verma et al., 2014), Gini Index (Alderighi et al., 2007; Scholz, 2012), and Hubbing Concentration Index (Martín and Voltes-Dorta, 2009). More specifically, clustering coefficient measures how complete the direct flight routes of an airport has in its neighbourhood. A high average clustering coefficient is often associated with a lower average travel distance and a fewer number of legs (Ponton et al., 2013). This is similarly implied from the closeness centrality, which, however, offers broader measures of accessibility between all neighbourhood airports. Passenger flux estimates the number of passengers travelling a particular route based on the flight frequency proportion of the destination airport. The flux of passengers on a connection can sometimes be overestimated as it depends only on flight frequency of available onward connections from a destination airport, but not directly the flight frequency between two airports. Based on total seat capacity of an airport, our flow betweenness centrality measures the intensity as a potential of the passenger flow between two airports.

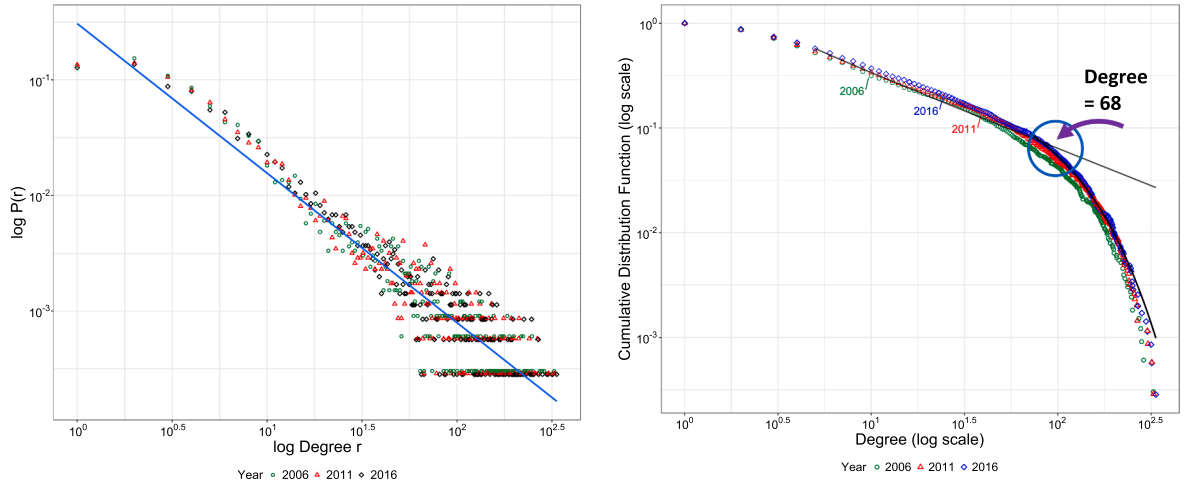
The other two measures are network concentration metrics which analyse routing through the network and the resulting traffic flows. Gini Index measures the spatial concentration of airline networks by calculating the traffic shares of all airports in the network, while Hubbing Concentration Index (HCI) measures the hubbing behaviour of the passengers connecting to their onward travels. However, both focus only on analysing the network structure of a single airline, rather than the whole of air transport network. Intuitively, the importance of an airport in hubbing and the passenger transfer traffic shares of that airport can be indirectly interpreted from our intensity-based flow betweenness and eigenvector centrality. Using our flow-betweenness measure, we estimated the proportions of transfer passengers at 11 hub-airports and compared to those actual data found in Lin and Zhang (2016). The result of our comparison is summarized in Table A1 of the Appendix. Our estimation is on average within 4% of the actual transfer percentages.

### 3. Individual metric results

In this section, we focus on the evolving changes of the five individual indicators over the 2006–2016 period, and the global passenger flows that are shaped by these changes.

#### 3.1. Degree centrality

The average degree in 2016 was 20.94, but around 70% of the airports had a degree of 13 or less. This was an increase of 17.4% compared to the average degree in 2006 - around 70% airports had a degree of 10 or less. The long right tail of these distributions suggests a power law-like distribution. This is typical of the air transport network in which most airports have a relatively small



**Fig. 2.** (a) Degree distribution log-log plot; and (b) Cumulative distribution function (CDF) of the global aviation network for years 2006, 2011 and 2016.

degree and a few hubs have a very large degree. The latter include Paris-CDG and Frankfurt-FRA that are connected by many airports. At a first glance, the node connection mechanism is governed by the rule called “preferential attachment” (Lordan et al., 2014; Sheridan and Onodera, 2018), which means that the more connected a node is, the greater chance it is to be connected by other nodes. However, the log-log plot of the degree distribution (see Fig. 2a) shows that the degree distribution does not actually follow the preferential-attachment rule well. The straight line fit of the log-log plot is largely arbitrary, as there are a lot of noise in the tail of the distribution. The plots of the empirical “cumulative distribution function” (CDF) for these three representative years (2006, 2011, and 2016) also indicate a faster than linear decay at the tail (Fig. 2b). The tail behaviour is better characterised by an exponential distribution. This suggests a power law decay combined with an exponential cut off (Clauset et al., 2009), which has a density function:

$$P(r) \propto r^{-\alpha} e^{-\beta r} \quad (7)$$

The power law scaling parameter  $\alpha$  is estimated at 1.49 with the cut-off degree at around 68, and the estimated rate parameter  $\beta = 0.0043$ . The degree is thus heavy-tailed with some airports being significantly larger than the average, but the air transport network is not scale-free. The total number of airports having degrees greater than the estimated cut-off degree of 68, increased from 225 in 2006, to 316 in 2016. This shows that medium to large airports enhanced their connectivity by opening direct routes to new destinations.

Another important feature we confirmed is that a high degree implies a high node weight. Over the study period, the mega-degree ( $r \geq 200$ ) airports achieved a compound passenger-capacity growth of 5.54% year on year, as compared to only 1.55% from those smaller airports with passenger capacity less than 10 million.

### 3.2. Flow betweenness

The results of computing  $F_b(i)$  for all the airports by their geographical regions in 2006 and 2016 are shown in Fig. 3. The betweenness centrality was plotted against their degrees and volumes. An airport with high flow betweenness is more likely to be located on the lowest “resistance” path between multiple origin-destination pairs in the network, and therefore more air traffic passes through it. The Sydney Airport (SYD) and Singapore Airport (SIN) were consistently the top two in betweenness centrality during the 10-years period. For many airports, high degree and/or high passenger capacity do not necessarily have high flow betweenness centrality. For example, the Atlanta International Airport (ATL) had a degree of 269 and a total passenger capacity of 134 M (M for million) in 2016, which are about 2 times the degree (136) and the passenger capacity (70 M) of the Kuala Lumpur International Airport (KUL) in Malaysia, and yet, KUL had more than double the flow betweenness centrality of ATL. When considering their geographical regions, airports located in Europe (EU) and North America (NA) generally had higher degrees, but relatively lower betweenness centrality than airports located in Asia (AS) and South-West Pacific (SW). The Middle-East (ME) airports have risen in their flow betweenness significantly in 2016 (Fig. 3b). In particular, Istanbul Atatürk Airport (IST), King Abdulaziz International Airport (JED), Bali Ngurah Rai International Airport (DPS), Shanghai Pudong International Airport (PVG), Beijing Capital International Airport (PEK) and Dubai International Airport (DXB) rose at a faster pace than other airports in the top-30 ranked betweenness airports. Over the same period, Los Angeles International Airport (LAX), Bangkok International Airport (BKK), John F. Kennedy International Airport (JFK) and Frankfurt Airport (FRA) had dropped the most. From this indicator, we can see that the role of some traditional hub airports is threatened by new rising airports. For example, Hong Kong International Airport (HKG), its flow betweenness had dropped from 7600 in 2006 to around 6700 in 2016. It had been overtaken by Beijing (PEK) and Shanghai Pudong (PVG). In addition, HKG’s nearest neighbour, Guangzhou airport (CAN) had jumped 43 positions in the betweenness ranking to reach

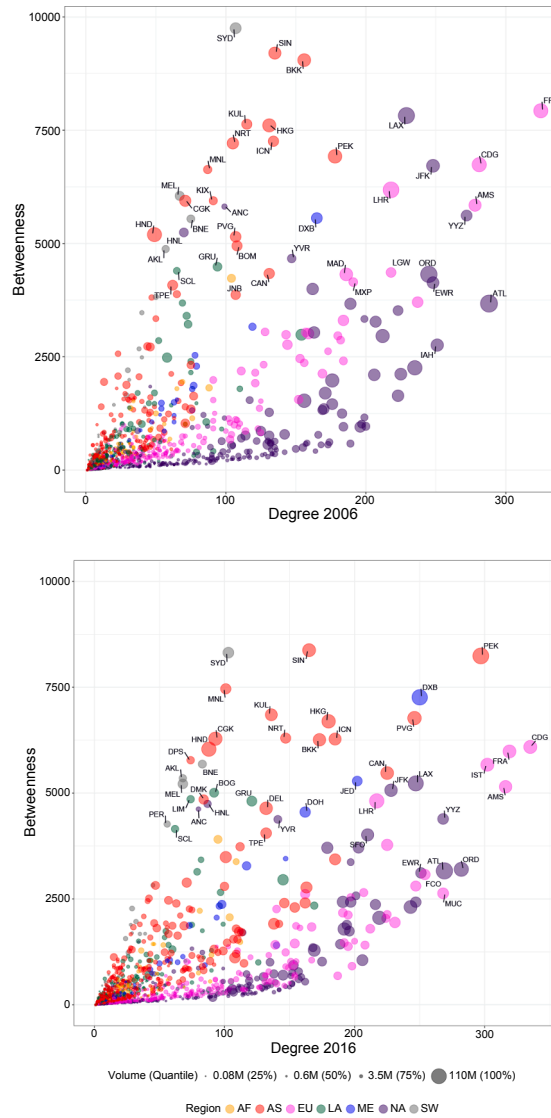


Fig. 3. Flow Betweenness vs. Degree in: (a) 2006 (Top); and (b) 2016 (Bottom).

within 5 positions off HKG over the same period. As these China airports have many route destinations overlapped with HKG, the transit-ability of HKG had been under threat, and more passengers chose to use those China airports as their gateway.

### 3.3. Closeness centrality

The two airports in Europe (EU), Paris Charles de Gaulle (CDG) and Frankfurt (FRA), were consistently ranked the top two in closeness centrality during the study period. Their mean path lengths were 2.19 and 2.23 respectively in 2016, compared with the longest mean path length of 5.14 in the air transport network. Compared to 4.4 in 2000 (Guimerà et al., 2005) and 4.37 in 2002 (Barrat et al., 2004), the world's mean path length of 3.54 in 2016 indicates, as expected, that the efficiency of air travel is being improved with better, shorter and more direct paths. The values of closeness centrality lie on a very small range, from 0.194 to 0.456 in 2016, 0.178–0.454 in 2011, and 0.158–0.451 in 2006. This means that the orders of closeness centrality are very sensitive to the change of network topology. In general, the mean path lengths between nodes in air transport network are typically small, where the longest path length was decreasing logarithmically with the size of the network. It is apparent that high degree airports generally have high closeness (Fig. 4). This is also evident by the work done by Yang and Zhou (2019) on China's airfreight airport-network, Ciliberto et al. (2019) on the US airport-network, and Wang et al. (2011) on China's airport city network. They found that the rankings by degree and closeness are generally consistent in their respective regions of study. However, the closeness centrality values can span a rather wide range for the same degree level. If an airport can strategically connect to a set of important airports (hubs), its closeness can be significantly improved. Another property of closeness centrality is its positive correlation (about 0.73)

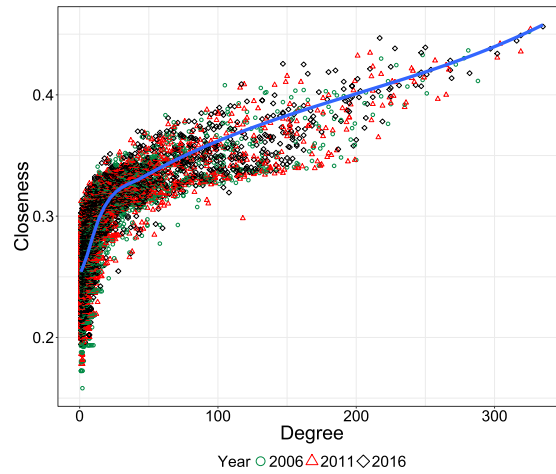


Fig. 4. Degree vs. Closeness in 2006, 2011 and 2016.

with the total capacity volume.

### 3.4. Eigenvector centrality

Many of the mega-degree airports in North America have been consistently ranked in the top eigenvector centrality, such as Los Angeles-LAX, Atlanta-ATL and Chicago-ORD. In contrast, many of the mega-degree airports in EU, such as Paris-CDG, Frankfurt-FRA, Amsterdam-AMS, and Istanbul-IST, had relatively average eigenvector centrality (Fig. 5). The strong presence of low-cost carriers at these airports have contributed to their average eigenvector centrality, as some of the airports to which they are connected are relatively low in the airport hierarchy. The only exception was London Heathrow (LHR), which was the only airport in Europe ranked in the top 10 in 2016. The airports in Asia region (AS) rose at a faster pace than most airports in the other regions. For example, Hong Kong-HKG, Beijing-PEK, Shanghai-PVG, Singapore-SIN, and Incheon-ICN have all reached the top 20 for the first time in 2016. Fig. 6 gives a summary of the eigenvector centrality distribution across the regions by 2006, 2011 and 2016 respectively. Furthermore, the eigenvector centrality appears to increase with the degree within a region because airports with low degree are connected mainly to airports with high degree and, as a result, have greater importance. This is supported also by the disassortative property of the air transport network and the fact that most connections are intra-regional. The positive rank-correlation between eigenvector and degree is echoed in the work done by Campante and Yanagizawa-Drott (2018), who reported a positive correlation between the number of international connections to cities and the eigenvector centrality of the city.

### 3.5. Regional importance

Different from the previous four centrality indicators, the regional importance is clearly not correlated well to the degree (Fig. 7). Though most small airports had very small regional importance, there were some exceptions. Comparing 2016 with 2006, the

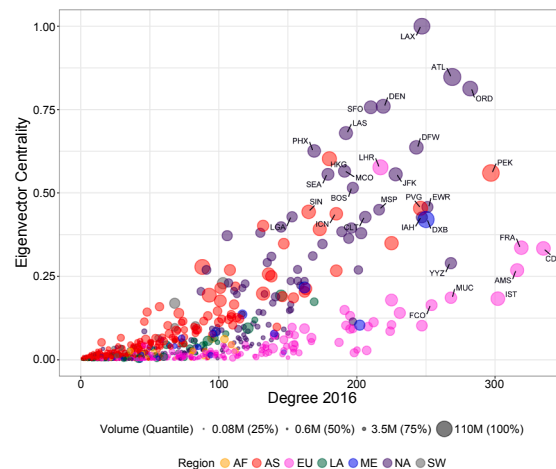


Fig. 5. Weighted Eigenvector Centrality vs. Degree 2016.

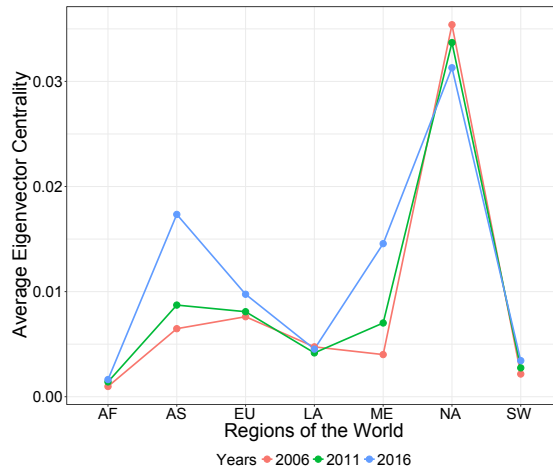


Fig. 6. Average Eigenvector Centrality Across 7 Regions of the World in 2006, 2011 and 2016.

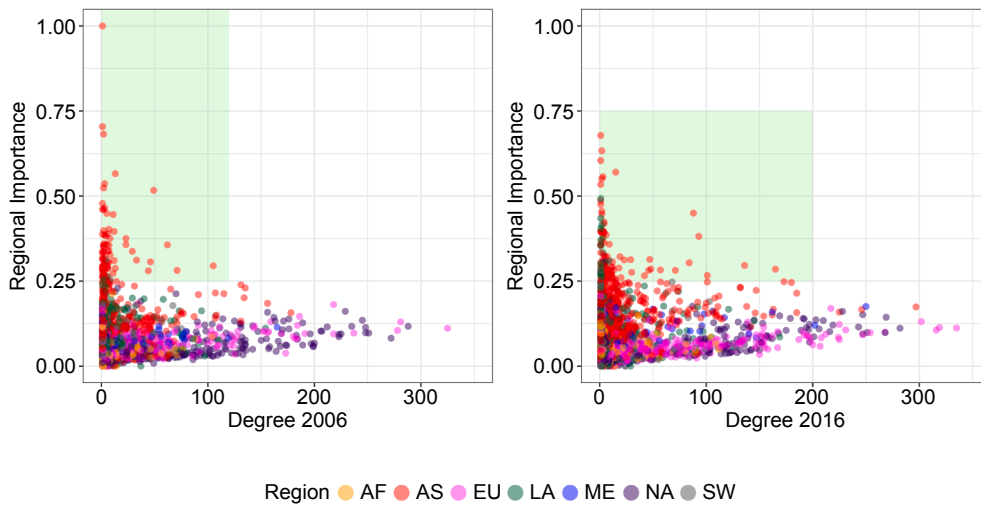


Fig. 7. Regional Importance vs. Degree: 2006 and 2016.

regional importance among airports of various degrees showed good consistency. The average  $C_R(i)$  in 2006 was 0.0621 and increased to 0.0740 in 2016.

We observed two changes here. First, the top regional importance in 2006, Yamaguchi Ube Airport (UBJ) dropped to the 6th position in 2016, with its  $C_R(i)$  dropped by almost half, to 0.551. This hinted that it was difficult to maintain very high capacity for only one route. Second, more medium-size airports expanded their regional connections. In 2006, there were only 78 airports with high  $C_R(i) \geq 0.25$ , and this had since increased to 125 airports in 2016. Interestingly, these airports of high regional importance were almost all located in Asia in 2006 but had scattered to a few other regions in 2016.

#### 4. Global airport connectivity index (GACI) evolution 2006–2016

One of the basic questions about air transport network is the identification of important airports worldwide. As global passenger flows become more concentrated in the hub cities and at the same time, more dispersed to secondary airports within those cities (Wong et al., 2019a), some secondary airports have risen in their global positions in the air transport network. While hub airports remain more connected than ever, emerging markets are gaining share of total inbound and outbound travellers. Some of the main airports in emerging markets have started to surpass the traditional hubs in global importance. We have developed a Global Airport Connectivity Index (GACI), which measures each airport's relative importance in the air transport network, by combining the five selected centrality indicators into a single score. It helps identify and assess dominant airports as to where they rank in relation to their counterparts, upon changes of network topology and passenger flows through the network over time. The weight associated with each indicator is tuned in a specific way which (i) retains the most valuable parts of all of the indicators, and (ii) gauges the potential scale of future passenger flows to facilitate comparison of GACI over time.

To create GACI, we first examine the correlations between the rankings obtained using the different centrality indicators and provide some guidance for the tuning of the weights. Specifically, we examine the four centrality indicators: degree, closeness, eigenvector, and flow-betweenness, which are node-level indices of centrality related more to the topology of the network. Their Spearman correlations are computed after ranking the indices. Scatter plot matrix of the four ranked indicators and their corresponding Spearman correlations are shown in Fig. A1 in the Appendix. The four indicators are clearly rank-correlated (with Spearman correlation  $> 0.7$ ), but this result does not diminish the fact that each is measuring a distinct aspect of the air transport network. Furthermore, the poor correlations appearing at lower ranks hint that these four indicators lack the capacity to rank small degree airports in the same order.

An additional characteristic of the air transport network has been investigated in an attempt to produce a more stable GACI ranking across all airport sizes, not just for the most highly-ranked nodes. The regional importance indicator  $C_R(i)$  is found to be an important quality metric of airport connections. To justify the choice made, we compare the correlations between GACI and capacity volumes across the four quantiles of the airport degrees, with and without  $C_R(i)$  included in the GACI. For the lower two quantiles, the correlation has improved from 0.3 to 0.51 when  $C_R(i)$  is included. At the third quantile, the improvement is similar, from 0.4 to 0.56. At the fourth quantile (largest degree), the improvement is substantially less, from 0.9 to 0.91 only, indicating that  $C_R(i)$  is less a contributing factor for mega-degree airports.

The GACI is computed using Principal Component Analysis (PCA) to extract the most informative component from the five centrality indicators. Of the number of multivariate methods available, PCA (Costa and Cesar, 2001; Duda et al., 2001) and Canonical Variate Analysis (CVA) (Duda et al., 2001; McLachlan, 2004) are the most popular approaches for exploring range of variation and interrelationships present in the entire dataset. While both methods are linear dimension reduction technique, they differ in the way the projection directions are found. In PCA, linear combinations of the original variables are found which capture most of the variance in the data and are orthogonal, whereas in CVA, they are selected to maximise the distances between groups relatively to the within group variance. In essence, CVA maximises the discrimination instead of variability of the data. However, the identification of a priori “group” for CVA presents a methodological problem. Thus, the PCA is preferred in this study in the absence of any priori decision of the group structure underlying the dataset.

The GACI is defined as:

$$\text{GACI} = [D, C_c, F_b, C_E, C_R] \alpha \quad (8)$$

where  $D, C_c, F_b, C_E, C_R$  correspond to the  $N$ -dimensional column vectors and  $\alpha = [\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5]$  is a weight vector determined by the first principal component or the direction of maximum variance in PCA. As these five indicators all have different ranges, raw scores of each indicator are standardised in a way that their ranking order within the same year and relative scores across the years are preserved. This makes GACI a time-independent index able to track and compare airport importance across time, both against the airports themselves and with each other. The weight vector  $\alpha$  for each of the air transport networks in 2006, 2011 and 2016 is shown in Table 2. The percentage of the total variance explained by the first principal component is found to be between 65% and 68%.

Table 3 shows the GACI changes over the 11-year period. Overall, majority of the airports had a marginal increase of GACI. However, there were still over 3100 airports, or 88% of the total, which had the GACI below 1.0 in 2016. As a reference for the range, airports with GACI above 1.0 had contributed to 81% of the total capacity. This confirms the observation that air transport network is highly concentrated at a few large airports (about the top 12% of the airports). Clearly, it is strategically important for airports to increase GACI as well as improve the ranking, by improving their underperformed indicators.

From the plot of GACI versus capacity volume of world airports in 2016 (Fig. 8), an airport with high GACI generally has high capacity volume. Apart from its country's and the world's GDP growth, an airport's capacity growth also depends on its centrality in the air transport network. Indeed, most airports' capacities have grown more quickly with higher GACI. In addition to the measurements related to direct connectivity and regional importance, it is also possible to extend GACI by directly including the capacity volumes between pairs of airports. Instead, we have characterized capacity volumes in a more generalized fashion in GACI, by considering the flow intensity of connections around each airport based on its node-strength (Eq. (3)), allowing the consideration of passenger spillover effects to the airport's connected neighbours in the presence of direct and indirect connections.

Another interesting observation from Fig. 8 is that the average capacity volume of airports in Asia had a highest volume per GACI when the GACI was below 2.75. Past this point, the AS airports were outperformed by those mega-degree airports in ME and NA, where there were higher traffic concentrations at larger hub airports. With the exception of LHR, most airports in Europe had lower capacity volume per GACI. In 2016, there were 1450 airports with GACI below 0.6. These airports generally had very low passenger volumes.

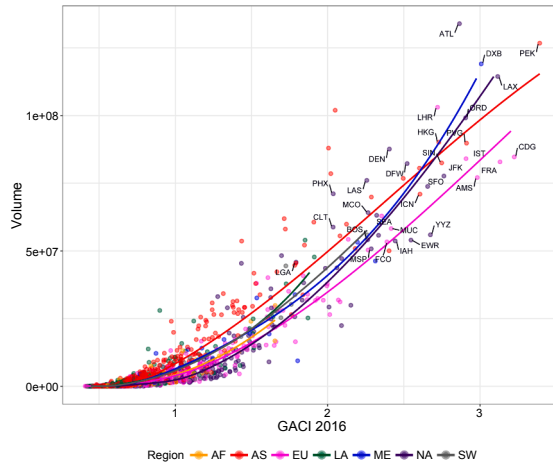
Table 4 shows the top-55 airports (with  $\text{GACI} \geq 2.0$ ) in 2016, and their GACI and rank changes since 2006. When considering

**Table 2**  
The Weight Vector and % of Variance Explained by the First Principal Component.

| Year | $D$   | $C_c$ | $F_b$ | $C_E$ | $C_R$ | % variance explained |
|------|-------|-------|-------|-------|-------|----------------------|
| 2006 | 0.675 | 0.494 | 0.435 | 0.312 | 0.118 | 65%                  |
| 2011 | 0.696 | 0.469 | 0.451 | 0.285 | 0.107 | 66%                  |
| 2016 | 0.704 | 0.458 | 0.439 | 0.297 | 0.115 | 68%                  |

**Table 3**  
GACI Evolution, 2006–2016.

| Year | Average GACI | Min GACI | Max GACI | Number of airports with GACI |         |         |       |
|------|--------------|----------|----------|------------------------------|---------|---------|-------|
|      |              |          |          | < 0.5                        | 0.5–1.0 | 1.0–2.0 | ≥ 2.0 |
| 2006 | 0.728        | 0.360    | 3.275    | 239                          | 2710    | 301     | 41    |
| 2011 | 0.719        | 0.380    | 3.286    | 345                          | 2731    | 333     | 49    |
| 2016 | 0.727        | 0.408    | 3.392    | 307                          | 2797    | 351     | 55    |



**Fig. 8.** 2016 GACI vs. Volume.

their geographical regions, NA had most airports (42%) in the top-55 GACI, followed by AS (27%) and EU (24%). Airports in AS and ME achieved the largest improvements in both GACI and ranks in the 11-year period. On average, ME airports had improved their GACIs by 47% and ranks by 59 positions. AS airports had improved their GACIs by 26% and ranks by 26 positions over the same period. On the other hand, many mega-degree airports in North America and Europe, like Washington-IAD, Detroit-DTW, Boston-BOS, New York-JFK, Frankfurt-FRA, and London-LHR performed worse than others. This shows that there was a shift of connectivity concentration from Europe and America to Asia and Middle East. It is important to note that NA and EU together have 40% more seat capacity than the whole of AS. Thus, airports in AS are more in competition with other EU and NA destinations than their intra-region counterparts. Furthermore, travel within Asia (AS3-AS4), and between EU1 and ME1, have greater growth potential. The former has almost doubled the share percentage of global flows from 5.4% to 10% between 2006 and 2016, while the latter has increased from 4.2% to 6.3%. In the same period, the busy transatlantic route (EU1-NA1) has dropped from 11% to 8.4%. The ability for large international airlines to serve wider Asian markets is most likely expedited through strategic partnerships and alliances with carriers having geographic positioning in Asia Pacific and possessing large fleets. As a consequence, some of the regional hub airports in NA and EU may establish more direct flights to their countries' gateway airports to serve the new opportunities in the high-growth Asian market area.

## 5. Findings, discussions and applications

### 5.1. GACI and global hub development

The five centrality indicators of GACI cover both structural and volumetric properties, allowing GACI as an effective classifier for hub airports. The classification of hub airports within each world region can also be determined by applying the concept of community in network science to the whole region. Some airports are connected within their region-as-a-community, and some have links evenly distributed among the regions. The measure of an airport's connectivity to its outside regions is called "participation coefficient" (Vargas and Wahl, 2014) in network analysis. Airports with high participation coefficients play a role of regional connector in the global air transport network. The participation coefficient for an airport  $i$  is defined as:

$$P(i) = 1 - \sum_{r=1}^R \left( \frac{d_{ir}}{d_i} \right)^2 \quad (9)$$

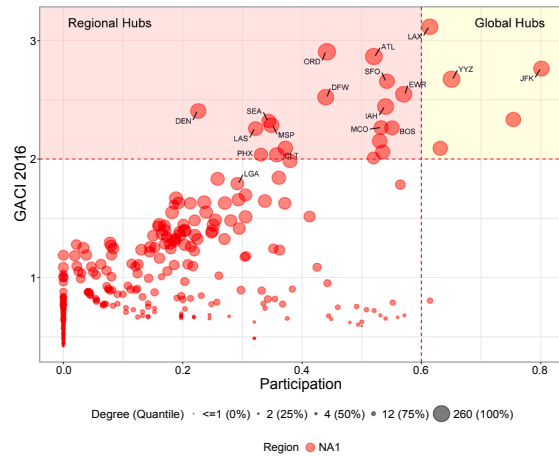
where  $R$  is the number of disjoint regions in the air transport network,  $d_{ir}$  corresponds to the number of links of airport  $i$  to other airports within region  $r$ , and  $d_i$  is the total degree of this airport.  $P(i) \approx 1$  if the links of airport  $i$  are uniformly distributed among the regions, and  $P(i) \approx 0$  if all of its links are within its own region. The number of disjoint regions,  $R$ , is based on the OAG's regional

**Table 4**  
Top-55 Airports Ranked by GACI, 2016.

| Airport (ranked by 2016 GACI) | GACI 2016 | GACI 2011 | GACI 2006 | Rank Change from 2006 | % GACI Change from 2006 | Region | Airport City-Country   |
|-------------------------------|-----------|-----------|-----------|-----------------------|-------------------------|--------|------------------------|
| PEK                           | 3.392     | 2.996     | 2.386     | 18                    | 42.14%                  | AS     | Beijing, China         |
| CDG                           | 3.225     | 3.083     | 2.942     | 3                     | 9.62%                   | EU     | Paris, France          |
| FRA                           | 3.131     | 3.286     | 3.275     | -2                    | -4.39%                  | EU     | Frankfurt, Germany     |
| LAX                           | 3.116     | 3.168     | 3.239     | -2                    | -3.80%                  | NA     | Los Angeles, USA       |
| DXB                           | 3.008     | 2.581     | 2.142     | 28                    | 40.40%                  | ME     | Dubai, UAE             |
| AMS                           | 2.982     | 3.031     | 2.779     | 1                     | 7.29%                   | EU     | Amsterdam, Netherlands |
| PVG                           | 2.912     | 2.423     | 1.918     | 39                    | 51.76%                  | AS     | Shanghai, China        |
| IST                           | 2.908     | 2.204     | 1.825     | 50                    | 59.41%                  | EU     | Istanbul, Turkey       |
| ORD                           | 2.906     | 2.860     | 2.950     | -5                    | -1.49%                  | NA     | Chicago, USA           |
| ATL                           | 2.865     | 3.167     | 3.023     | -7                    | -5.23%                  | NA     | Atlanta, USA           |
| JFK                           | 2.763     | 2.679     | 2.905     | -5                    | -4.89%                  | NA     | New York, USA          |
| SIN                           | 2.748     | 2.862     | 2.470     | 3                     | 11.26%                  | AS     | Singapore              |
| HKG                           | 2.730     | 2.466     | 2.391     | 5                     | 14.18%                  | AS     | Hong Kong              |
| LHR                           | 2.721     | 2.640     | 2.751     | -5                    | -1.09%                  | EU     | London, UK             |
| YYZ                           | 2.674     | 2.652     | 2.766     | -7                    | -3.32%                  | NA     | Toronto, Canada        |
| SFO                           | 2.656     | 2.609     | 2.321     | 4                     | 14.46%                  | NA     | San Francisco, USA     |
| ICN                           | 2.606     | 2.387     | 2.274     | 7                     | 14.59%                  | AS     | Seoul, Korea           |
| CAN                           | 2.603     | 2.265     | 1.817     | 43                    | 43.23%                  | AS     | Guangzhou, China       |
| EWR                           | 2.547     | 2.569     | 2.649     | -9                    | -3.87%                  | NA     | New York, USA          |
| DFW                           | 2.522     | 2.648     | 2.552     | -9                    | -1.18%                  | NA     | Dallas, USA            |
| BKK                           | 2.496     | 2.599     | 2.522     | -9                    | -1.03%                  | AS     | Bangkok, Thailand      |
| IAH                           | 2.443     | 2.433     | 2.503     | -9                    | -2.40%                  | NA     | Houston, USA           |
| MUC                           | 2.417     | 2.337     | 2.310     | -1                    | 4.64%                   | EU     | Munich, Germany        |
| DEN                           | 2.405     | 2.714     | 2.485     | -10                   | -3.22%                  | NA     | Denver, USA            |
| NRT                           | 2.403     | 2.217     | 2.262     | 2                     | 6.22%                   | AS     | Tokyo, Japan           |
| FCO                           | 2.389     | 2.312     | 2.038     | 13                    | 17.22%                  | EU     | Rome, Italy            |
| MAD                           | 2.354     | 2.389     | 2.158     | 4                     | 9.10%                   | EU     | Madrid, Spain          |
| MIA                           | 2.334     | 2.251     | 2.212     | 2                     | 5.53%                   | NA     | Miami, USA             |
| SEA                           | 2.322     | 2.447     | 2.158     | 3                     | 7.57%                   | NA     | Seattle, USA           |
| JED                           | 2.313     | 1.850     | 1.634     | 55                    | 41.58%                  | ME     | Jeddah, Saudi Arabia   |
| KUL                           | 2.287     | 2.417     | 2.134     | 3                     | 7.17%                   | AS     | Kuala Lumpur, Malaysia |
| MSP                           | 2.286     | 2.304     | 2.237     | -3                    | 2.18%                   | NA     | Minneapolis, USA       |
| MCO                           | 2.266     | 2.191     | 2.091     | 3                     | 8.33%                   | NA     | Orlando, USA           |
| LGW                           | 2.265     | 2.194     | 2.263     | -8                    | 0.08%                   | EU     | London, UK             |
| BOS                           | 2.263     | 2.334     | 2.399     | -19                   | -5.67%                  | NA     | Boston, USA            |
| SYD                           | 2.260     | 2.299     | 2.292     | -13                   | -1.43%                  | SW     | Sydney, Australia      |
| LAS                           | 2.257     | 2.440     | 2.270     | -12                   | -0.58%                  | NA     | Las Vegas, USA         |
| DOH                           | 2.196     | 1.778     | 1.386     | 93                    | 58.40%                  | ME     | Doha, Qatar            |
| MNL                           | 2.181     | 2.090     | 1.863     | 15                    | 17.09%                  | AS     | Manila, Philippines    |
| IAD                           | 2.152     | 2.452     | 2.399     | -23                   | -10.27%                 | NA     | Washington, USA        |
| BCN                           | 2.135     | 2.187     | 1.881     | 8                     | 13.50%                  | EU     | Barcelona, Spain       |
| CTU                           | 2.124     | 1.526     | 1.258     | 131                   | 68.87%                  | AS     | Chengdu, China         |
| DTW                           | 2.092     | 2.274     | 2.321     | -22                   | -9.85%                  | NA     | Detroit, USA           |
| YUL                           | 2.091     | 2.108     | 2.042     | -6                    | 2.38%                   | NA     | Montreal, Canada       |
| MAN                           | 2.082     | 1.891     | 1.934     | -1                    | 7.64%                   | EU     | Manchester, UK         |
| TPE                           | 2.081     | 1.864     | 1.708     | 23                    | 21.80%                  | AS     | Taipei, China          |
| BRU                           | 2.070     | 2.166     | 1.770     | 18                    | 16.96%                  | EU     | Brussels, Belgium      |
| FLL                           | 2.061     | 1.956     | 1.700     | 26                    | 21.26%                  | NA     | Fort Lauderdale, USA   |
| HND                           | 2.049     | 1.882     | 1.449     | 64                    | 41.41%                  | AS     | Tokyo, Japan           |
| PHX                           | 2.037     | 2.115     | 2.086     | -13                   | -2.36%                  | NA     | Phoenix, USA           |
| CLT                           | 2.036     | 1.965     | 1.942     | -9                    | 4.84%                   | NA     | Charlotte, USA         |
| DEL                           | 2.021     | 1.888     | 1.696     | 24                    | 19.16%                  | AS     | Delhi, India           |
| MXP                           | 2.014     | 2.020     | 2.116     | -16                   | -4.82%                  | EU     | Milan, Italy           |
| YVR                           | 2.009     | 1.929     | 2.030     | -13                   | -1.03%                  | NA     | Vancouver, Canada      |
| CGK                           | 2.005     | 2.077     | 1.703     | 17                    | 17.73%                  | AS     | Jakarta, Indonesia     |

classification as shown in Appendix [Table A2](#).

The participation coefficient and GACI were examined together to quantify the airport hubness. A hub was defined with  $GACI \geq 2.0$ . This value was found by applying the optimal one-dimensional k-means clustering algorithm ([Wang and Song, 2011](#)) on the GACI in 2016. The algorithm essentially partitions the GACI values of 3510 airports into a number of groups such that the sum of squares within-groups is minimized. The  $GACI \geq 2.0$  corresponds to the smallest GACI of the group that has the highest group-mean. Airports in this group are sufficiently well-connected to be considered as hubs. These hub airports can be further classified into regional hubs or global hubs, depending on their participation coefficient. Following the work by [Guimerà et al. \(2005\)](#), nodes with participation coefficient ( $P \leq 0.62$ ) were classed as nodes with most links within their community, so-called “nonhub connector nodes”, while ( $P > 0.62$ ) were classified as nodes with links homogeneously distributed among all communities. Thus, we define a

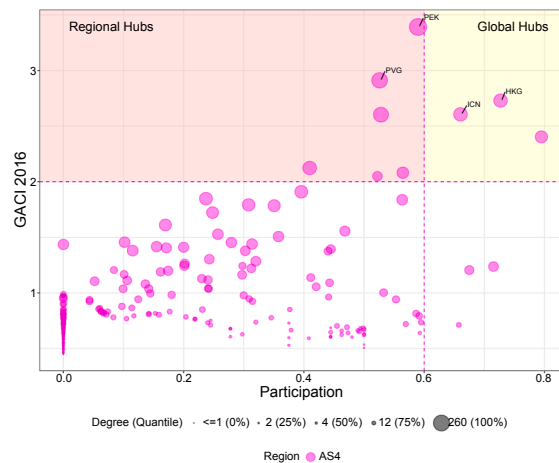


**Fig. 9.** a. NA1 Region - Global and Regional Hubs. b. AS4 Region – Global and Regional Hubs. c. EU1 Region – Global and Regional Hubs. d. ME1 Region – Global and Regional Hubs.

regional hub with  $P(i) \leq 0.6$  and a global hub with links evenly distributed among all 17 regions ( $P(i) > 0.6$ ). In total, there were 29 global hubs and 26 regional hubs in the world in 2016. There were less than 6% of the airports with the participation coefficients greater than 0.6. This suggests that most of the airports are connecting to their own regions. In 2016, there were over 75% of routes and over 80% airline capacities belong to intra-regional. Fig. 9 shows the results of GACI against participation coefficient in four main regions (EU1, NA1, AS4 and ME1). The hub airports in these regions have taken different pathways in their developments. For government policy makers, understanding and leveraging the position of their city airports is critical. This requires prompt decisions to be made about expanding the corresponding infrastructures and implementing suitable policy and incentive schemes to protect the airport's position. For instance, an airport located within short- and medium-haul reach of an international market can position itself as being a regional hub in a part of the low-cost carrier network, for connecting passengers between a hub city and the overseas market. Support from the city government to provide surface transport connections to the airport is deemed critical to enhance public accessibility (Zhu et al., 2018). Airports must also provide appropriate infrastructure such as sufficient passenger facilities and aircraft parking positions. Understanding the route opportunities that can improve the airport's GACI and matching the infrastructure to required aircraft types are just two possible strategic actions of the airport to attract more international airlines.

**North America, China and Russia focused on developing regional hubs:** Despite having the largest number of airports in its region, NA1 only had five global hubs, namely, Los Angeles-LAX, New York-JFK, Toronto-YYZ, Miami-MIA, and Montréal-YUL. In AS4, three airports were classified as global hubs: Hong Kong-HKG, Tokyo-NRT, and Seoul-ICN. Both these regions have quite many regional hubs. There was a rising trend that many of the larger-volume airports in these regions are getting close to the GACI 2.0 mark. They are expected to develop into regional hubs in the near future.

**West EU and Middle East focused on developing Global Hubs:** The mega airports in these two regions had developed themselves into global hubs. West Europe (EU1) had the largest number of global hubs (11), while ME1 had 3. These airports play a similar role in connecting a large number of passengers on transatlantic routes. Furthermore, ME1 had more airports in the lower right quadrant, indicating its efforts focusing more on expanding secondary airports into global hubs, instead of regional hubs.



**Fig. 9.** (continued)

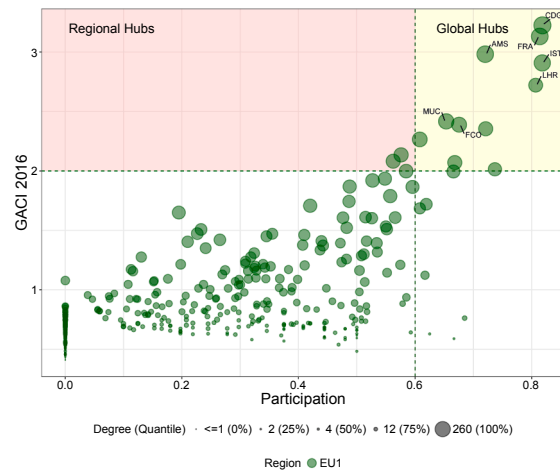


Fig. 9. (continued)

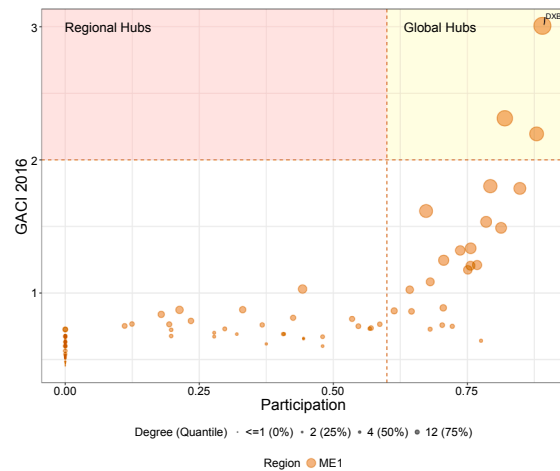


Fig. 9. (continued)

### 5.2. Major and secondary airports developments in different regions

Airports with GACI 1.0 – 2.0 (Table 3) are considered as the rising stars. These 300+ airports contributed to about 50% of the world passenger seat capacity. They are also mostly operated as major or secondary airports in their region and would potentially become the super hub airports in the future. Table 5 summarises the statistics of this group of airports across the seven regions of the world. Europe was the fastest growing region for this type of airports, and was followed by Asia. On the other hand, North America dropped significantly. Indicated also from the analysis of participation coefficients in Section 5.1, it is expected that some of these rising stars will become global hub airports in the future. More specifically, as indicated in Section 5.1 Europe has already had the largest number of global hubs. This hub market is consolidated and is difficult for airlines to enter due to capacity limitations caused

Table 5

Performance of Major or Secondary Airports (with GACI 1.0–2.0) across Different Regions.

| Region | 2006 GACI |        |                 |        | 2016 GACI |        |                 |        |
|--------|-----------|--------|-----------------|--------|-----------|--------|-----------------|--------|
|        | Average   | Median | No. of airports | Total  | Average   | Median | No. of airports | Total  |
| AF     | 1.18      | 1.10   | 14              | 16.46  | 1.22      | 1.13   | 17              | 20.73  |
| AS     | 1.24      | 1.11   | 51              | 63.03  | 1.29      | 1.21   | 77              | 98.98  |
| EU     | 1.30      | 1.20   | 76              | 98.49  | 1.31      | 1.22   | 108             | 141.70 |
| LA     | 1.27      | 1.21   | 27              | 34.41  | 1.30      | 1.21   | 32              | 41.63  |
| ME     | 1.23      | 1.19   | 16              | 19.67  | 1.35      | 1.28   | 14              | 18.86  |
| NA     | 1.34      | 1.29   | 107             | 143.78 | 1.32      | 1.28   | 92              | 121.70 |
| SW     | 1.28      | 1.18   | 10              | 12.83  | 1.28      | 1.13   | 11              | 14.13  |

by, among others, environmental concerns. Examples include the construction delay in the third runway at Heathrow airport, and the capacity limit at Schiphol airport (500,000 flights/year, until 2020). Schiphol already reaches the limit, and nobody knows for sure what will happen after 2020.<sup>1</sup> Different from Burghouwt (2007) who pointed out that major European airlines were still concentrated at a few large hub airports in the early 2000s, Bel and Fageda (2008), Reynolds-Feighan (2010), Suau-Sanchez and Burghouwt (2011) and Wong et al. (2019a) found a de-concentration and hub by-passing trend at some traditional European mega airport hubs during recent years. The tight global hub market would give opportunities for some major secondary airports to move up the ladder. At the same time, new connectivity strategies together with a traffic spill from the largest airports may result in more and new smaller hubs in Europe.<sup>2</sup>

While for Asia, there will be fast development of both global and regional hubs, led by the rapid growth of air services and the continuous rapid expansion of LCC networks – according to CAPA (2019), LCCs have quadrupled their total seat capacity in the region over the past decade. As mentioned in the introduction, the aviation industry has led to increased employment and service quality in local markets, especially for the high-tech, business, local tourism and logistics sectors. Some of those fast rising airports can leverage off established international passenger flows to their regions with connections to the higher GACI gateway airports, and build up sufficient frequency and capacity to enable commercially viable direct air services to those international destinations. To airlines, these are also the new routes and new bases that could be explored to increase connectivity. By strategic partnerships with the airlines, those fast rising airports and their networks can effectively feed and connect passengers to a larger network. Table 5 shows that many airports in Asia and Europe appear to have great growth potentials. For North America, 25 airports have their GACI dropped below 1.0, while only five airports manage their GACI to rise above 1.0, suggesting that the airport expansion in NA may slow down in the near future.

In terms of the individual indicator performance, the flow-betweenness, closeness and regional importance had minor improvements across the regions over the years for this category of airports. The degree and eigenvector centralities, however, demonstrated significant improvements, particularly in Asia and Middle East (Fig. 10). This shows that some of the airports in this category have been operating as good major or secondary airports, which improved, perhaps strategically, their GACI by connecting to more and important airports (super hubs) in the global network.

### 5.3. Development of new routes with high GACI

As shown in Table 4, there were several airports attained significant level of improvement in GACI over the 11-year period. This section takes some of the good and poor performers from the four main regions: Asia, Europe, Middle East and North America, and analyses their similarities. Specifically, the three good performers are PVG in Shanghai, IST Istanbul, DXB Dubai, and the poor performer IAD Washington DC. In addition, there appears to be no significant change in South West Pacific (SW1) Region. For the good performers, the number of new routes were all substantially more than their cancelled routes. The quality of those new routes measured by the GACI of their destination airports is at a high average (Table 6), compared to the global average GACI of 0.727 in 2016 (Table 3). On the other hand, the poor performer, IAD, experienced a significant negative change in rank and GACI. It had cancelled 77 routes but developed only 55 new routes during the same period. Unfortunately, the cancelled route quality is also at a high average, which contributes to IAD's drop in GACI ranking. Clearly, it is in the interest of the airport to recommend, promote, and perhaps incentivise new flights for a new route, particularly to those destinations that can improve the airport's own GACI, thereby elevating the importance of the city in the global air transport network.

### 5.4. GACI changes across 17 regions

By further comparing the GACI changes of all the airports in the 11-year period, there were total of 1192 airports experiencing absolute changes of 0.05 or above. The rest of the 1707 Airports, which had a GACI change less than 0.05 had an average net change of new and cancelled routes less than 2. The changes to their air transport network connectivity were thus insignificant. On the other hand, there were 32 airports with large absolute GACI changes ( $>0.4$ ), and 12 of them that had their GACI crossed above the 1.0 mark showed a significant improvement in connectivity to become a major or hub airport in their region in 2016. Table 7 shows the summary statistics of the airports and their new/cancelled routes across the 17 regions with different ranges of GACI changes.

Clearly, the new connectivity was not chosen at random. To achieve a positive GACI change, airports must strategically develop their new routes. Many airports actually preferred introducing new routes within their own region over inter-regions. However, it is often the case that inter-region new routes would have a larger (positive or negative) impact to the GACI changes. Among airports that have positive GACI changes, most of them are located in North Asia (mostly China in the AS4 region), followed by West Europe (EU1) and South East Asia (AS3). From the airport connectivity perspective, Asia and Europe are the two leading regions of fast

<sup>1</sup> The plan “Smart and Sustainable Flying” launched by the Dutch aviation industry in 2018 contains action points that would make flying less polluting, so that additional growth would be (perceived as) less harmful. Furthermore, the city of Amsterdam is actively trying to limit the number of tourists in the city centre (by introducing higher tourist taxes, limiting the number of buses, etc.).

<sup>2</sup> Outside of Europe, studies in the late 1990s and early 2000s concluded that traffic concentration in the form of “hub and spoke” network dominated the airline industry in the US (e.g., Goetz and Sutton, 1997; Reynolds-Feighan, 2001) and in Southeast Asia (Bowen, 2000). Later studies nevertheless suggest a more mixed outlook, or even a dispersal trend in global air travel (e.g., O'Connor, 2003; Reynolds-Feighan, 2010; Wong et al., 2019a, 2019b).

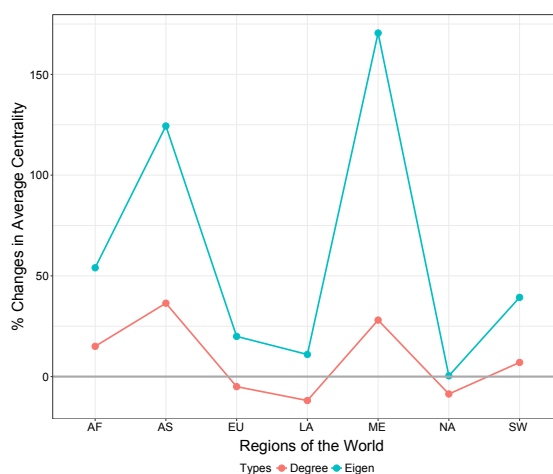


Fig. 10. Average Degree and Eigenvector Centrality Percentage Changes for Airports with GACI 1.0 – 2.0.

Table 6

Selected Airports Development in the Top Class (2016 vs. 2006).

| Airport (ranked by 2016 GACI) | GACI 2016 | GACI 2006 | Rank Change from 2006 | % GACI Change from 2006 | No. of new routes | No. of cancelled routes | GACI of new destination airports Mean (min – max)          |
|-------------------------------|-----------|-----------|-----------------------|-------------------------|-------------------|-------------------------|------------------------------------------------------------|
| PVG                           | 2.912     | 1.918     | 39                    | 51.76%                  | 148               | 9                       | 1.02 (0.63–2.87)                                           |
| DXB                           | 3.008     | 2.142     | 28                    | 40.40%                  | 107               | 22                      | 1.32 (0.67–3.12)                                           |
| IST                           | 2.908     | 1.825     | 50                    | 59.41%                  | 161               | 16                      | 1.05 (0.62–3.12)                                           |
| IAD                           | 2.152     | 2.399     | –23                   | –10.27%                 | 55                | 77                      | GACI of cancelled destination airports<br>1.07 (0.65–2.52) |

Table 7

Summary Statistics of GACI Changes among 17 Regions, 2006 – 2016.

| GACI Change (+ ve)                                              | > 0.4  | 0.2–0.4 | 0.05–0.2 | GACI Change (– ve)              | –0.05 to –0.2 | –0.2 to –0.4 | < –0.4 |
|-----------------------------------------------------------------|--------|---------|----------|---------------------------------|---------------|--------------|--------|
| Number of airports                                              | 32     | 101     | 515      | Number of airports              | 500           | 42           | 2      |
| Avg GACI 06                                                     | 1.2296 | 1.1552  | 0.7826   | Avg GACI 06                     | 0.8334        | 1.2798       | 1.6106 |
| Avg GACI 16                                                     | 1.8626 | 1.4288  | 0.8890   | Avg GACI 16                     | 0.7430        | 1.022        | 1.1359 |
| Avg no. of net routes                                           | 80.25  | 43.772  | 15.2136  | Avg no. of net routes           | –5.564        | –36.929      | –52    |
| Avg Max new destination GACI                                    | 3.0580 | 2.9087  | 2.6381   | Avg Max cancel destination GACI | 2.1735        | 2.7341       | 3.2754 |
| Avg Min new destination GACI                                    | 0.6000 | 0.60778 | 0.7247   | Avg Min cancel destination GACI | 0.8091        | 0.6579       | 0.6079 |
| Regional new routes                                             | 1562   | 3115    | 6671     | Regional cancel routes          | 4930          | 1868         | 83     |
| Inter-region new routes                                         | 1227   | 2389    | 3842     | Inter-region cancel routes      | 1468          | 313          | 42     |
| <b>Airport Region Distribution with Significant GACI Change</b> |        |         |          |                                 |               |              |        |
| AF1                                                             | 0      | 3       | 16       | AF1                             | 15            | 1            | 0      |
| AF2                                                             | 0      | 1       | 3        | AF2                             | 7             | 0            | 0      |
| AF3                                                             | 0      | 1       | 10       | AF3                             | 13            | 0            | 0      |
| AF4                                                             | 1      | 0       | 11       | AF4                             | 5             | 0            | 0      |
| AS1                                                             | 0      | 4       | 22       | AS1                             | 18            | 1            | 0      |
| AS2                                                             | 0      | 0       | 10       | AS2                             | 4             | 0            | 0      |
| AS3                                                             | 2      | 10      | 46       | AS3                             | 19            | 0            | 0      |
| AS4                                                             | 19     | 22      | 108      | AS4                             | 28            | 0            | 0      |
| NA1                                                             | 0      | 4       | 43       | NA1                             | 203           | 37           | 1      |
| LA1                                                             | 0      | 1       | 12       | LA1                             | 21            | 0            | 0      |
| LA2                                                             | 0      | 2       | 9        | LA2                             | 20            | 0            | 0      |
| LA3                                                             | 0      | 2       | 10       | LA3                             | 13            | 0            | 0      |
| LA4                                                             | 0      | 3       | 30       | LA4                             | 14            | 2            | 0      |
| EU1                                                             | 5      | 35      | 109      | EU1                             | 48            | 0            | 0      |
| EU2                                                             | 0      | 8       | 47       | EU2                             | 17            | 0            | 0      |
| ME1                                                             | 5      | 5       | 19       | ME1                             | 25            | 1            | 1      |
| SW1                                                             | 0      | 0       | 10       | SW1                             | 30            | 0            | 0      |

growth. In terms of capacity changes, the average growth percentage per airport that has a positive GACI change over the 11-year period is 963%, which is more than 3 times the global average of 293%.

Among airports with negative GACI changes, most of them had significant decreases in the net number of routes. Many of these airports also had a high proportion of cancelled routes with relatively higher GACI airports. They were mostly located in North America (NA1), indicating a diminishing trend in global connectivity by the US airports during the 11 years. In terms of capacity changes, the average growth percentage for this group of airports that have negative GACI changes is merely 7.6%, which is well below the global average.

## 6. Concluding remarks

This paper has analysed the nature and evolution of the world air transport network by developing and applying a Global Airport Connectivity Index (GACI). This GACI has incorporated the topology and volumetric characteristics of an airport, as well as its regional importance in the global air transport network. More specifically, the GACI combines existing centrality indicators including degree, closeness and eigenvector, together with two new volume-based indicators: flow betweenness and regional importance. The flow-betweenness measures the importance of an airport as the transit point on connection-paths between pairs of origin and destination. Regional importance measures the link intensity of an airport's connections within its region. Together, these five centrality indicators captured a broad range of air transport network specific properties, such as power law degree distribution, hubness, preferred within-region connections, faster growth in passenger capacity of larger airports, short mean path length, logarithmical shrinkage in network size over time, and distinct regional patterns of global passenger flows. By computing the GACI changes over the years, we can evaluate the significance of network developments due to changes induced by new connections to the airports.

The analysis of the historical seat capacity data of 3500 airports, together with the airports' complex network structure, revealed deeper insights in the developments of air transport network over time. First, the air transport network is not scale-free. Majority of the capacity growth concentrated at the mega-degree ( $r \geq 200$ ) airports, which achieved passenger capacity compound growth about four times faster than those smaller airports. An airport with a high closeness centrality score tends to have a high capacity volume. From the flow-betweenness performance, we found that most Middle-East and Chinese airports were on the rise and some of them have overtaken the traditional hubs in Europe and North America over this study period. Instead of using the traditional US and European airports for transit, a large number of passengers switched to use Middle-East and Asian airports. Lastly, with the exception of North America, all the other regions had improved their eigenvector centrality scores, with Middle-East and Asia airports had more than doubled their average scores over the same period. These airports are getting very prominent in the air transport network.

The GACI was calculated for every airport, and the evolution of air transport networks across all the world regions were studied. First, different regions appeared to have different strategies in expanding their air transport networks. Both America and North Asia focused on developing regional hub airports, serving short- to medium-haul routes to address regional demands. While for West Europe, South East Asia and Middle East, they seemed to prefer to developing their major airports into global hubs, serving long-haul and international flights. Although the average GACI did not change much during this period, there was a fast growth in European and Asia secondary airports. To maintain or improve the hub position of an airport with large traffic volume, the airport must develop new routes connecting to higher GACI airports.

As airport competes with one another on passenger traffics and service quality to both airlines and passengers (Wiltshire, 2018), the public utility status of airports has been fading and airports have been increasingly being operated like a business. Many airports are under growing pressure to be more financially self-sufficient (e.g. Zhang and Zhang, 2003). For the regulators, they should find ways to optimise the allocation of additional routes in a liberalising airline market, which can maximise social welfare and minimise passengers' travel disutility (Li et al., 2010). To many travellers, airport and route choice is always a joint decision (Yang et al., 2014), especially with long-haul journey and transit flights. These dynamic changes in aviation business environment encouraged the vertical airport–airline relationships, in order to strengthen the hub status of major airports (Zhang and Czerny, 2012; Yang et al., 2015). Many airports and airlines established close and long-term relationship to attain a win-win situation: with airports being able to attract more passengers from existing (with higher frequency) and new routes (direct flights), substantial profits can be generated from the terminal, including those from leasing areas used for catering and retailing. The findings in this paper shed some lights to policy makers and industry leaders in planning their aviation strategies.

In the future, the application of GACI on airports and routes development can be further explored, particularly in the areas of airport classification and strategic network change configuration pertinent to airport growth. A GACI-based airport classification can add the connectivity dimension to many existing airport classification schemes that are often based on such factors as public access, runway length, commercial services, and general aviation facilities. It can also precisely define the airport roles, such as super-hub, major, regional, local and minor airport, in the regional and global aviation network. Furthermore, GACI can provide a measure of the network changes to specific airports, assisting policy makers to understand, for instance, the effects of strategy route changes by other airports. The performance of an airport is affected by not only its local and regional economy, but also its connectivity to other airports, as well as other airports' connectivity. Being able to capture, examine and predict the temporal changes of the air transport network, more insightful analyses can be performed so as to improve the competitiveness of an airport city. Indeed, a similar GACI can be also computed at an airport-city level to measure the connectivity and importance of an airport-city in the global air transport network. This would represent a natural extension of the analysis presented in the present paper.

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## Appendix A

**Table A1**

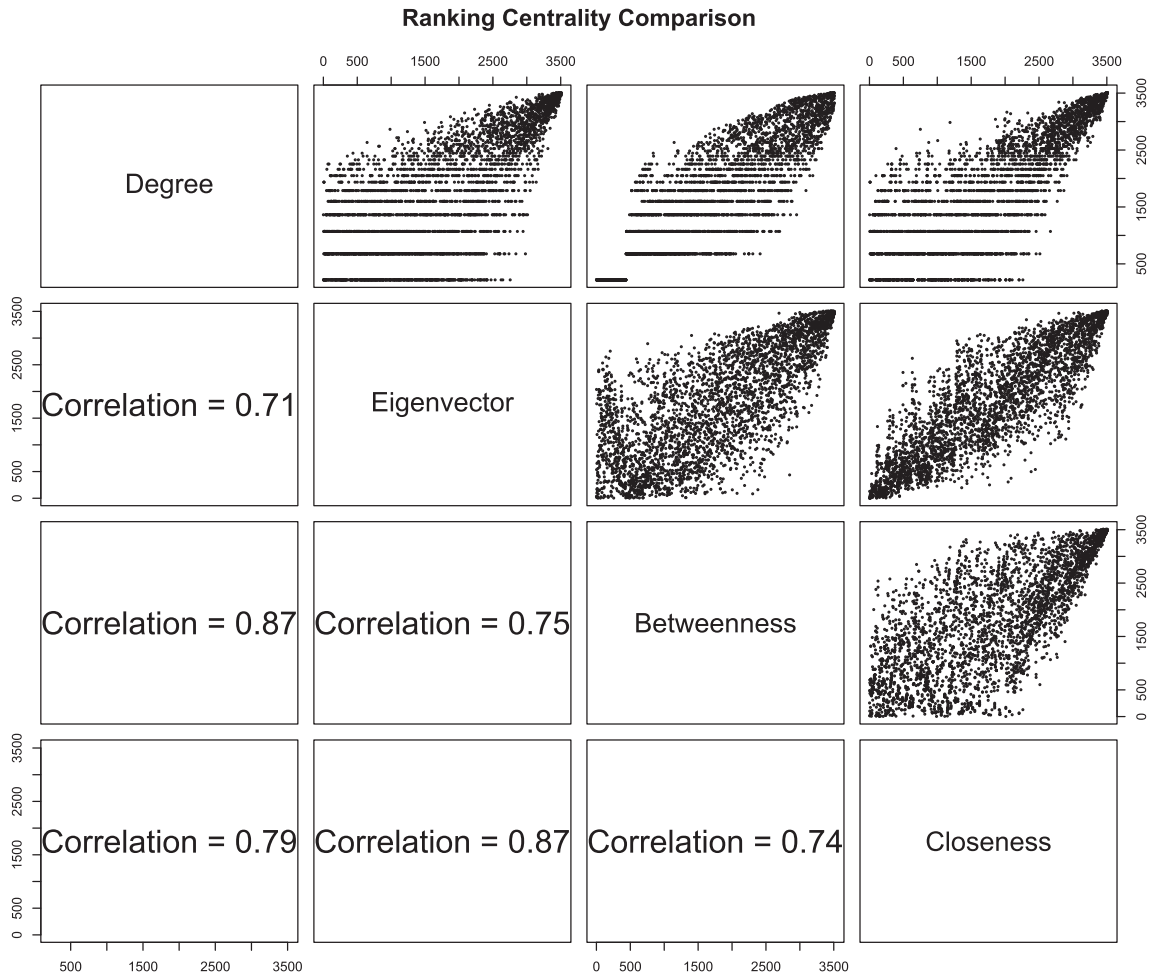
The proportion of transfer passengers at major airports.

| Data from Lin and Zhang (2016) |                                 |            |      | Transfer % estimated from Intensity-based Betweenness | Year |
|--------------------------------|---------------------------------|------------|------|-------------------------------------------------------|------|
| Airport Code                   | Airport Name                    | Percentage | Year |                                                       |      |
| FRA                            | Frankfurt                       | 54         | 2007 | 57.8                                                  | 2011 |
| CDG                            | Paris Charles de Gaulle         | 52         | 2011 | 50.1                                                  | 2011 |
| DXB                            | Dubai Int'l                     | 52         | 2016 | 61                                                    | 2016 |
| AMS                            | Amsterdam Schiphol              | 41.9       | 2013 | 49.7                                                  | 2013 |
| MUC                            | Munich, FranzJosef Stauss Int'l | 39         | 2013 | 26.7                                                  | 2013 |
| LHR                            | London Heathrow                 | 37         | 2013 | 36.4                                                  | 2013 |
| SIN                            | Singapore Changi Int'l          | 30         | 2013 | 40.3                                                  | 2013 |
| HKG                            | Hong Kong Int'l                 | 26         | 2013 | 26.5                                                  | 2013 |
| CPH                            | Copenhagen Kastrup              | 20.8       | 2014 | 21.1                                                  | 2014 |
| ICN                            | Incheon Int'l                   | 19         | 2013 | 21.7                                                  | 2013 |
| NRT                            | Narita Int'l                    | 18.5       | 2013 | 15.9                                                  | 2013 |

**Table A2**

OAG regional definitions and the top GACI airports.

| OAG Region Code | Region                 | No. of Airports | Top GACI Airport 2016 |                            |
|-----------------|------------------------|-----------------|-----------------------|----------------------------|
| AF1             | North Africa           | 90              | CAI                   | Cairo, Egypt               |
| AF2             | Southern Africa        | 76              | JNB                   | Johannesburg, South Africa |
| AF3             | Central/Western Africa | 101             | LOS                   | Lagos, Nigeria             |
| AF4             | Eastern Africa         | 90              | ADD                   | Addis Ababa, Ethiopia      |
| AS1             | South Asia             | 152             | DEL                   | Delhi, India               |
| AS2             | Central Asia           | 45              | TAS                   | Tashkent, Uzbekistan       |
| AS3             | South East Asia        | 271             | SIN                   | Singapore, Singapore       |
| AS4             | North East Asia        | 393             | PEK                   | Beijing, China             |
| NA1             | North America          | 759             | LAX                   | Los Angeles, USA           |
| LA1             | Caribbean              | 91              | SJU                   | San Juan, Puerto Rico      |
| LA2             | Central America        | 103             | CUN                   | Cancun, Mexico             |
| LA3             | Upper South America    | 120             | BOG                   | Bogota, Colombia           |
| LA4             | Lower South America    | 168             | GRU                   | Sao Paulo, Brazil          |
| EU1             | Western Europe         | 473             | CDG                   | Paris, France              |
| EU2             | Eastern/Central Europe | 181             | DME                   | Moscow, Russia             |
| ME1             | Middle East            | 107             | DXB                   | Dubai, UAE                 |
| SW1             | Southwest Pacific      | 290             | SYD                   | Sydney, Australia          |



**Fig. A1.** The Correlation Matrix Plot of Four Centrality Indicators (Degree, Weighted Eigenvector, Flow Betweenness and Closeness).

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